Dynamic Optimal Pricing with Love of Variety

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Introduction

- Repeated consumption may reduce the appeal of a product.
- Purchase histories can reveal when a consumer is ready to switch.
- A retailer can use that information to optimally price.

Research question

How do preferences for variety change the firm's optimal pricing strategy?

Motivation and Contribution

Motivation

- **Pricing simulation:** knowing switching types raises per-unit profits by up to about 2 cents in the reported normalization.
- **Demand:** flavor preferences change little when fatigue is included; some fatigue parameters are weakly identified.

Contribution

- Study attribute switching within brands as well as across brands.
- Connect switching preferences and state dependence to the information a retailer uses for pricing.

Supply exercise

- Purchase histories can reveal preferences for switching between products.
- The supply exercise asks how a retailer would price if switching types were known or inferred.
- Two products, two types, two periods.

Supply model

Consumer's value from choosing product j

$$V_{rjt} = \underbrace{\beta_{0j}}_{\text{baseline appeal}} + \underbrace{\gamma_r \mathbf{1}\{j \neq j_{t-1}\}}_{\text{reward for switching}} + \underbrace{\alpha p_{rjt}}_{\text{price effect}}, \qquad r \in \{H, L\}.$$

- V_{rjt} is the deterministic utility of product j for a type-r consumer at time t.
- Switching from the previously chosen product j_{t-1} adds γ_r; choosing it again adds no switching reward.
- High types value switching more: $\gamma_H > \gamma_L$. A higher price lowers utility ($\alpha < 0$).
- Consumers choose the highest utility after logit choice shocks; the outside option allows no purchase.

Full information

$W_r(j)$ is the retailer's optimal period-two profit per consumer, given type r and first-period choice j.

$$W_r(j) = \max_{\mathbf{p}_2(j)} \sum_k s_{rk2}(j)\big(p_{rk2}(j) - c_k\big),$$

$$\Pi^{\text{full}} = \sum_r N_r \sum_j s_{rj1}\left[p_{rj1} - c_j + \delta W_r(j)\right].$$

Perfect information

The retailer conditions prices on type and purchase history.

A lower price today can move a consumer into a more profitable state tomorrow.

Learning

$$q = \Pr(r = H), \qquad \tilde{s}_{j1} = q s_{Hj1} + (1 - q) s_{Lj1},$$

$$q_2(j) = \Pr(H \mid j) = \frac{q s_{Hj1}}{q s_{Hj1} + (1 - q) s_{Lj1}}.$$

- Period-one prices are common across unobserved types.
- Period-two prices depend on observed purchase history.
- A choice relatively more likely for high types raises the posterior.

Limited information

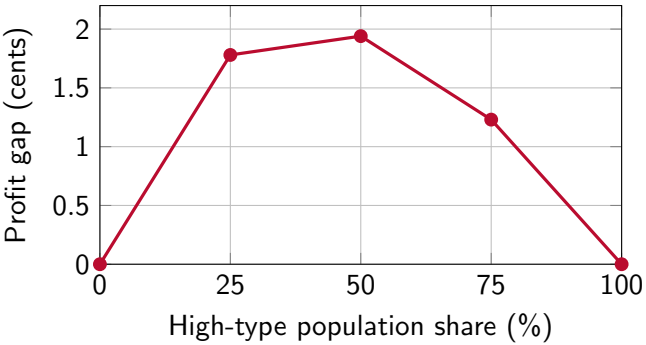
$$\Pi^{\text{limited}} = N \sum_j \tilde{s}_{j1}(\mathbf{p}_1) \left[p_{j1} - c_j + \delta W\big(j, q_2(j; \mathbf{p}_1)\big) \right].$$

1. Current margin versus current demand.
2. How purchases change future consumption states.
3. How purchases reveal switching type.

Mechanism

Prices influence both allocation and the informativeness of observed choices.

Supply results



Reported value of information

Full-information profit minus limited-information profit.

Largest gap: 1.94 cents at a 50/50 type mix.

Simulation parameters

Pricing results

Information Structure, Optimal Pricing, and Value of Information Across Population Mixes

Population Mix	Imperfect Price (P_1)		Posterior Beliefs		Imperfect	Perfect	Value of
	Good 1	Good 2	G1	G2	Profit	Profit	Info
100% LOW	2.0633	1.8646	0.0000	0.0000	0.6031	0.6031	-0.0000
25% HIGH, 75% LOW	2.1847	1.9944	0.3727	0.3641	0.7820	0.7999	0.0178
50% HIGH, 50% LOW	2.2819	2.1049	0.6510	0.6448	0.9773	0.9966	0.0194
75% HIGH, 25% LOW	2.3542	2.1887	0.8528	0.8504	1.1811	1.1934	0.0123
100% HIGH	2.4067	2.2494	1.0000	1.0000	1.3902	1.3902	0.0000

Data

- Nielsen HomeScan and MarketScan, 2022–2024.
- Eight markets: Atlanta, Chicago, Denver, Houston, Des Moines, Phoenix, Philadelphia, San Diego.
- Single-person households reduce ambiguity about whose tastes drive purchases.
- Single-serving yogurt offers flavor variation and limited storability.

Sample

	Full sample	Estimation sample
Households	5,408	998
Mean store trips / household	308.15	400.05
Mean observed yogurt choice weeks	—	3.91
Coupon / deal purchase rate	26.52%	23.10%
Mean weekly yogurt units (purchasers)	—	8.44

Implication

The panel contains limited yogurt history for many households.

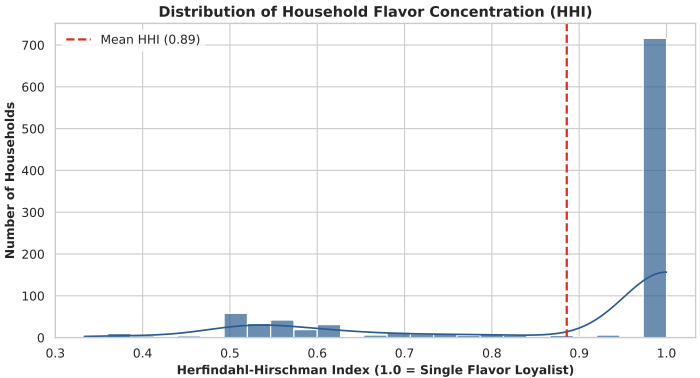
Flavor groups

- **Berry:** berry-labeled flavors plus cherry.
- **Plain:** a separate category.
- **Other:** all remaining flavors.
- **Outside option:** no yogurt purchase.

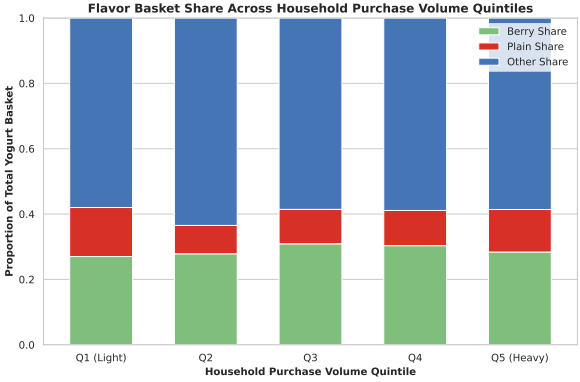
Tradeoff

Aggregation makes estimation tractable but hides switching within a flavor group.

Flavor concentration



Flavor shares



Demand model

$$U_{ijt} = \underbrace{\beta_0 + \beta_i' X_{jt}}_{\text{baseline tastes}} + \underbrace{\gamma_i \Xi_{ijt}}_{\text{history state}} + \underbrace{\alpha p_{jt}}_{\text{price}} + \varepsilon_{ijt}.$$

- Inside choices: berry, plain, other; outside utility normalized to zero.
- Ξ_{ijt} summarizes prior consumption and switching.
- Negative fatigue coefficients reduce utility as the history index rises.

Simulation procedure Simulation results Utility specification

Identification

- **Stable tastes:** persistent differences in flavor choices.
- **Fatigue:** changes in choices associated with a household's own history.
- **Challenge:** heterogeneity, prices, and unobserved shocks can also generate switching.
- **Limited history and coarse groups** make the distinction difficult.

Price control function

$$p_{jmt} = a + b \underbrace{\bar{p}_{j,-m,t}}_{\text{other-market price}} + \tau_t + \eta_{jmt}, \qquad \hat{\eta}_{jmt} = p_{jmt} - \hat{p}_{jmt}.$$

1. Match household choices to flavor-specific prices and purchase histories.
2. Regress local prices on other-market prices and time controls.
3. Include the fitted residual in utility to account for endogenous prices.

Required exclusion

Other-market prices must shift local prices without sharing unobserved local demand shocks, conditional on the controls.

Specifications

	No fatigue	With fatigue
Standard logit	Fixed β; $\gamma = 0$	Fixed β, γ
Random coefficient logit	Heterogeneous β; $\gamma = 0$	Heterogeneous β, γ

- Price and the control-function residual enter all four specifications.
- Compare baseline flavor tastes and price effects across columns.
- Assess fatigue means and dispersion separately from optimizer convergence.
- Identification depends on within-household history variation, valid price instruments, and the treatment of initial histories.

Panel likelihood

Demand estimates

	Standard logit		Random coefficients	
	No fatigue	Fatigue	No fatigue	Fatigue
Berry preference	−.7635	−.7576	−.7632	−.7489
Plain preference	−1.3899	−1.3897	−1.3930	−1.3956
Price	−.9836	−.9821	−.9836	−.9804
Control function	.1670	.1667	.1654	.1662

Main empirical result

The largest baseline flavor change is about 1.9% for berry in the random coefficient model.

LOV coefficients

	Standard logit	Random coefficients
Berry fatigue	−.7262	−7.2030
Standard error	(.1485)	(1.5424)
Plain fatigue	−.0351	−.0555
Standard error	(.1009)	(.1032)

- Berry has a negative estimated history coefficient.
- Plain fatigue is not statistically distinguishable from zero.
- Large changes across specifications warrant caution.

Heterogeneity

Random coefficient SD	Estimate	SE
β_{berry}	1.0513	.0916
β_{plain}	1.0513	.1492
γ_{berry}	148.4132	155.3868
γ_{plain}	1.0513	.1641

Diagnostic

The berry fatigue dispersion is extremely large and imprecisely estimated.

Optimization success alone does not establish reliable identification.

Implied WTP

Magnitude in dollars	Standard	Random coefficients
Berry preference	.77	.76
Plain preference	1.42	1.42
Berry fatigue	.74	7.35
Plain fatigue	.04	.06

$$\text{Implied WTP} = \frac{|\text{utility coefficient}|}{|\alpha|}.$$

Interpretation

The large \$7.35 number belongs to berry fatigue and is sensitive to the specification.

Policy discussion

- Restricting type information can reduce retailer profits in this model.
- The loss is greatest when the population contains both switching types.
- Attribute fatigue lowers consumer utility
- Difficult to disentangle preferences from love of variety

Simulation parameters

Baseline product utilities	$(2.2, 2.0)$
Switching utility (γ_H, γ_L)	$(2.0, 0.1)$
Price coefficient α	-1.5
Discount factor δ	0.95
Unit costs (c_1, c_2)	$(1.2, 1.0)$
Initial product-state weights	$(0.5, 0.5)$
Price bounds	$[\$1, \$8]$

Comparison

Vary the high-type population share under full and limited information.

◂ Supply results

Simulation procedure

Simulation design	Assigned value
Replications / evaluation periods / burn-in	1,000 / 100 / 5
Product attributes / baseline taste	$X_j \in \{1, 2, 3, 4, 5\}$; $\beta = 2$
Variety grid / choice shocks	$\gamma \in \{0, 3, 6, 9\}$; $\varepsilon_{jt} \sim \mathrm{T1EV}$

1. For each γ, initialize consumption histories during burn-in.
2. At each evaluation date, combine baseline attribute utility with the history-dependent distance reward and a choice shock.
3. Record the selected product, update its consumption history, and repeat.
4. Summarize choice probabilities across the simulated runs.

Purpose: show how variety changes choices in a controlled setting.

Simulation results ◂ Demand model

Simulation results

	$\gamma = 0$	$\gamma = 3$	$\gamma = 6$	$\gamma = 9$
Product 1 ($X = 1$)	.0016	.1352	.3321	.3973
Product 2 ($X = 2$)	.0084	.1325	.0541	.0084
Product 3 ($X = 3$)	.0437	.0844	.0026	.0000
Product 4 ($X = 4$)	.1934	.0553	.0023	.0002
Product 5 ($X = 5$)	.7529	.5926	.6089	.5941

Interpretation

Without variety, the highest-attribute product dominates. With stronger variety, choices concentrate at the endpoints.

This motivates modeling history; it does not establish identification in the yogurt panel.

Simulation procedure ◂ Demand model

Utility specification

For $j \in \{B, P, O\}$ (berry, plain, other), let $D_j^f = \mathbf{1}\{j = f\}$:

$$
\begin{aligned}
U_{ijt} &= V_{ijt} + \varepsilon_{ijt}, \\
V_{ijt} &= \beta_0 + \beta_i^B D_j^B + \beta_i^P D_j^P \\
&\quad + \gamma_i^B D_j^B \Xi_{ijt} + \gamma_i^P D_j^P \Xi_{ijt} + \alpha p_{j,m(i),t} + \sigma \hat{\eta}_{j,m(i),t}.
\end{aligned}
$$

- Other flavor is the reference; outside deterministic utility is zero.
- Ξ_{ijt} summarizes own past consumption and alternative-flavor history.
- History weight $\lambda = 0.7$ and decay $\delta = 0.9$ are fixed.
- Random coefficients allow persistent household differences in tastes and fatigue.

◂ Demand model

Panel likelihood

Conditional on household coefficients $b_i = (\beta_i^B, \beta_i^P, \gamma_i^B, \gamma_i^P)$:

$$P_{ijt}(b_i) = \frac{e^{V_{ijt}(b_i)}}{1 + \sum_{k \in \{B,P,O\}} e^{V_{ikt}(b_i)}}, \qquad P_{i0t}(b_i) = \frac{1}{1 + \sum_k e^{V_{ikt}(b_i)}}.$$

$$L_i(b_i) = \prod_{t \in \mathcal{T}_i} \prod_{j \in \{0,B,P,O\}} P_{ijt}(b_i)^{y_{ijt}}, \qquad y_{ijt} = \mathbf{1}\{\text{observed choice is } j\}.$$

$$\boxed{\hat{\theta} = \arg\max_\theta \sum_i \log\left[\int L_i(b)\, dF(b;\theta)\right]}$$

Simulation approximation: $\int L_i(b) dF(b;\theta) \approx R^{-1} \sum_{r=1}^R L_i(b_i^{(r)})$.

Maximize log-likelihood: $\ell(\theta) = \sum_{i,t,j} y_{ijt} \log P_{ijt}(\theta)$.

◂ Specifications

Discussion

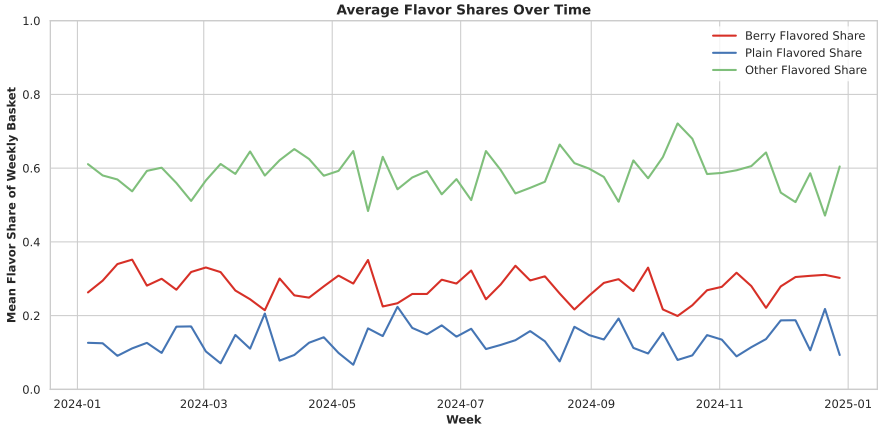
Questions for discussion

How can fatigue be distinguished from persistent unobserved tastes?

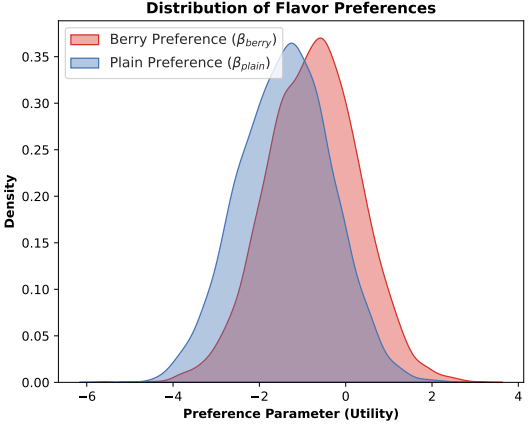
Which purchase histories are most informative about switching types?

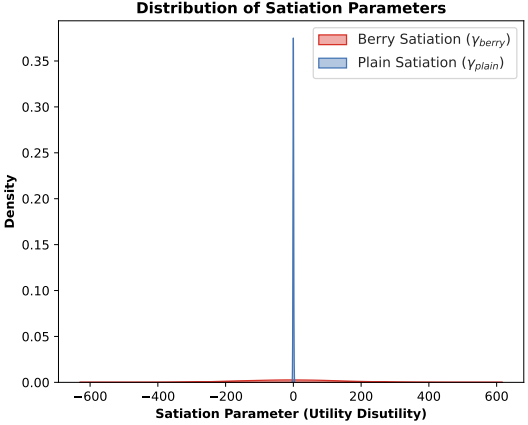
What consumer-welfare evidence is needed to evaluate privacy restrictions?

Flavor shares over time

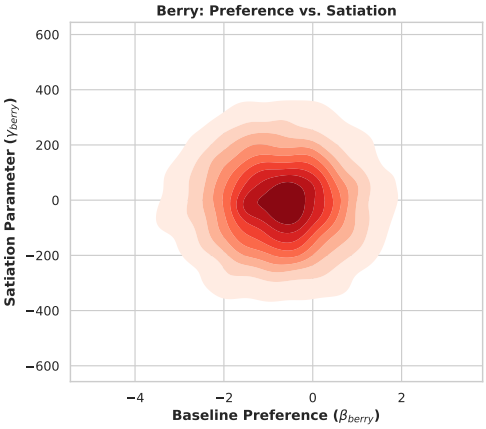


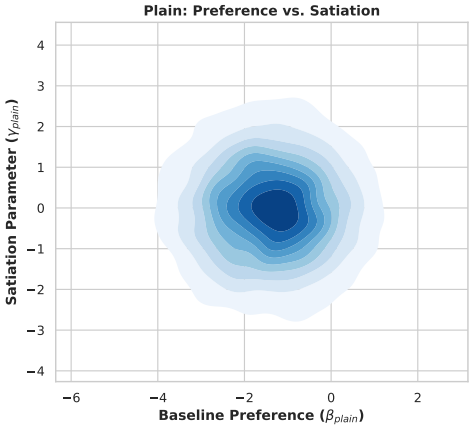
Marginal distributions





Joint distributions





References

- Attribute satiation: McAlister; Lattin (1987).
- Variety and brand choice: Givon (1984); Chintagunta (1999).
- Bundles and variety: Dubé (2004).
- Price control functions: Petrin and Train (2010).
- Personalized pricing: Dubé and Misra (2023).

Full bibliographic entries appear in the accompanying paper.